

Task Summary: Recover Graphviz Model Lineage

Think of this project as a "**digital detective**" task for an AI agent.

Imagine you have a messy, incomplete ledger (a CSV file) that keeps track of how a machine learning model was trained. Some important information is missing—like which model "parented" another, how many labels were used, and exactly which version of the dataset was involved.

This project tests if an AI can fix that mess. Here is the simple breakdown of what the AI has to do:

- **The Setup:** We give the AI a "corrupted" list and a pre-downloaded dataset (the `dair-ai/emotion` dataset) to work with offline, meaning it cannot cheat by searching the internet.
- **The Tools:** The AI is tasked with using two main tools: **Terraform** (to manage the process and orchestrate the steps) and **Python** (to do the heavy lifting of reading the data and performing the repairs).
- **The Goal:** The AI must write a script that fills in the missing information, sorts the records properly, and creates a visual map (a lineage diagram) showing how the different models are related.
- **The Proof:** Finally, we have an automated test (the "Verifier") that checks the AI's work. It looks for specific things, like:
 - Is the ledger fixed and organized correctly?
 - Does the visual diagram (SVG) look right?
 - Are the calculations for the labels accurate?

Conclusion: In short, we built a standardized test to see if an AI is smart enough to clean up messy data and turn it into a clear, reliable report without needing any outside help. It passed all tests with a perfect score.